

ALLIANCE UNIVERSITY

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NAAC GRADE A+ ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No: 5

Title: Building Customer Analysis Dashboard

GITHUB LINK

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Experiment No. 5: Building Customer Analysis Dashboard

1. Problem Statement

Businesses need to understand customer purchasing behaviour, sales performance, and revenue trends to make better decisions. Analysing customer transactions helps identify popular product categories, preferred payment methods, high-value customers, and revenue concentration. The objective of this experiment is to develop an interactive Customer Analysis Dashboard in Microsoft Power BI and communicate business insights through appropriate visualizations.

2. Objectives

1. Identify the product category generating the most revenue and orders.
2. Determine the most frequently used payment method.
3. Analyse how revenue changes month by month during 2025.
4. Identify customers with the highest number of purchases.
5. Evaluate whether a small group of customers accounts for a large share of revenue.
6. Present key performance indicators using interactive cards, charts, tables, and slicers.

3. Dataset Description

The analysis uses two CSV datasets imported into Power BI Desktop.

Table	Size	Main columns
Customers	1,000 rows	CustomerID, CustomerName, Gender, Age, City, State, IncomeBand, SignupDate, CustomerSegment
Orders	5,000 rows	OrderID, CustomerID, OrderDate, ProductCategory, Product, Quantity, UnitPrice, DiscountPercent, SalesAmount, PaymentMethod

The order dataset covers 1 January 2025 to 30 December 2025. It contains six product categories and five payment methods. A total of 916 unique customers placed at least one order.

4. Data Import and Preparation

- Imported both CSV files through Home → Get Data → Text/CSV.
- Checked data types: OrderDate and SignupDate were set as Date; Quantity, UnitPrice, DiscountPercent and SalesAmount were checked as numeric fields.
- Verified the key fields and ensured the OrderID values were unique.
- Created a one-to-many relationship from Customers[CustomerID] to Orders[CustomerID], with Customers on the one side.
- Used OrderDate by month on the monthly revenue trend chart and added slicers for product category, date, payment method and customer segment.

5. DAX Measures

The following measures were created in Power BI and used by the dashboard. Each formula should be added through Modeling → New measure. The model table containing order data is named Orders.

5.1 Total Revenue

```
Total Revenue =
SUM ( Orders[SalesAmount] )
```

5.2 Total Orders

```
Total Orders =
DISTINCTCOUNT ( Orders[OrderID] )
```

5.3 Active Customers

```
Active Customers =
DISTINCTCOUNT ( Orders[CustomerID] )
```

5.4 Average Order Value

```
Average Order Value =
DIVIDE ( [Total Revenue], [Total Orders], 0 )
```

5.5 Total Quantity

```
Total Quantity =
SUM ( Orders[Quantity] )
```

5.6 Top 10 Customer Revenue

```
Top 10 Customer Revenue =
VAR TopCustomers =
    TOPN ( 10, ALLSELECTED ( Orders[CustomerID] ), [Total Revenue], DESC )
RETURN
    SUMX ( TopCustomers, CALCULATE ( [Total Revenue] ) )
```

6. Dashboard Development

The report is presented on a single Power BI page. KPI cards are positioned along the left, slicers across the top, and charts and the customer table in the main canvas.

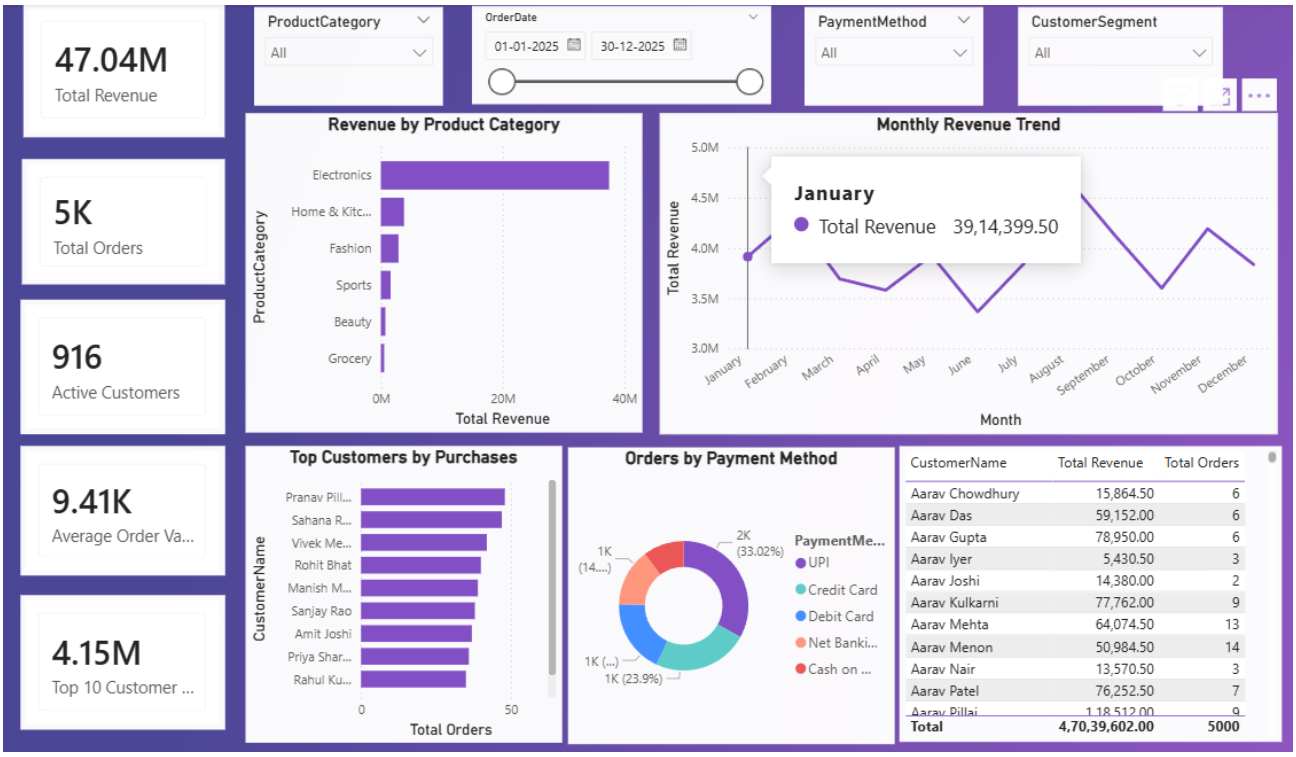


Figure 1. Customer Analysis Dashboard created in Power BI.

6.1 KPI Cards

KPI	Dashboard value	Meaning
Total Revenue	47.04M	Sum of SalesAmount across all orders.
Total Orders	5,000	Distinct count of OrderID.
Active Customers	916	Customers who placed at least one order out of 1,000 profiles.
Average Order Value	9.41K	Total Revenue divided by Total Orders.
Top 10 Customer Revenue	4.15M	Revenue from the ten highest-revenue CustomerIDs; approximately 8.81% of total revenue.

6.2 Slicers

Four slicers are included: ProductCategory, OrderDate, PaymentMethod and CustomerSegment. These let the viewer filter the report by the product category, date range, payment option or customer segment. Visuals respond to slicer selections according to their interaction settings.



6.3 Revenue by Product Category

A horizontal bar chart compares total revenue generated by each product category. Electronics is the clear leader in both revenue and order count.

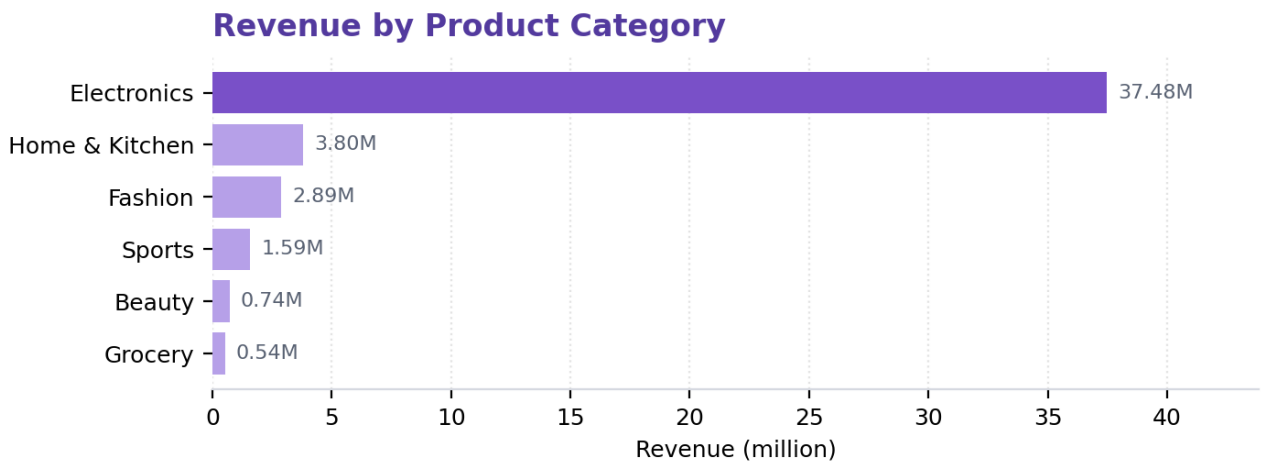


Figure 2. Revenue by product category.

Product category	Orders	Revenue	Share of revenue
Electronics	1,095	37.48M	79.67%
Home & Kitchen	867	3.80M	8.08%
Fashion	1,076	2.89M	6.15%
Sports	603	1.59M	3.38%
Beauty	672	0.74M	1.57%
Grocery	687	0.54M	1.14%

Observation

Electronics generated 37.48M from 1,095 orders, contributing 79.67% of total revenue. Home & Kitchen ranked second by revenue at 3.80M, followed by Fashion at 2.89M. Electronics therefore dominates the sales mix and represents a concentration risk if demand for that category weakens.

6.4 Monthly Revenue Trend

The line chart tracks monthly revenue from January to December 2025, showing changes in sales performance across the year.

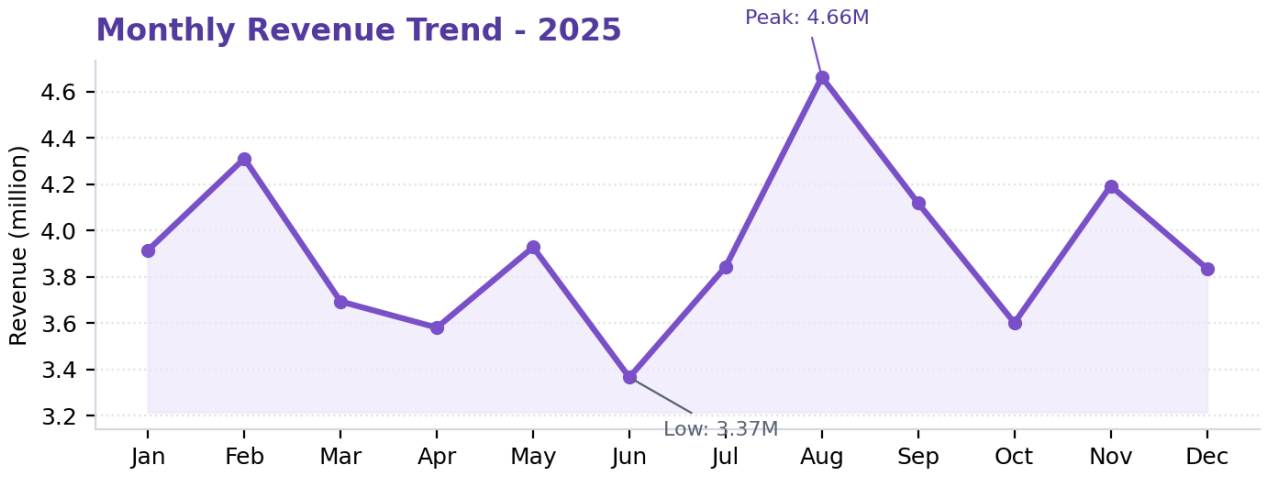


Figure 3. Monthly revenue trend, 2025.

Month	Revenue	Orde
January	3.91M	389
February	4.31M	402
March	3.69M	464
April	3.58M	401
May	3.93M	447
June	3.37M	388
July	3.84M	431
August	4.66M	446
September	4.12M	384
October	3.60M	421
November	4.19M	402
December	3.84M	425

Observation

Monthly revenue ranged from 3.37M in June to 4.66M in August. August recorded the highest revenue; June recorded the lowest. Average monthly revenue was approximately 3.92M. The trend fluctuates during the year but does not show a continuous upward or downward pattern.

6.5 Orders by Payment Method

The donut chart presents the number and percentage of orders by payment method.

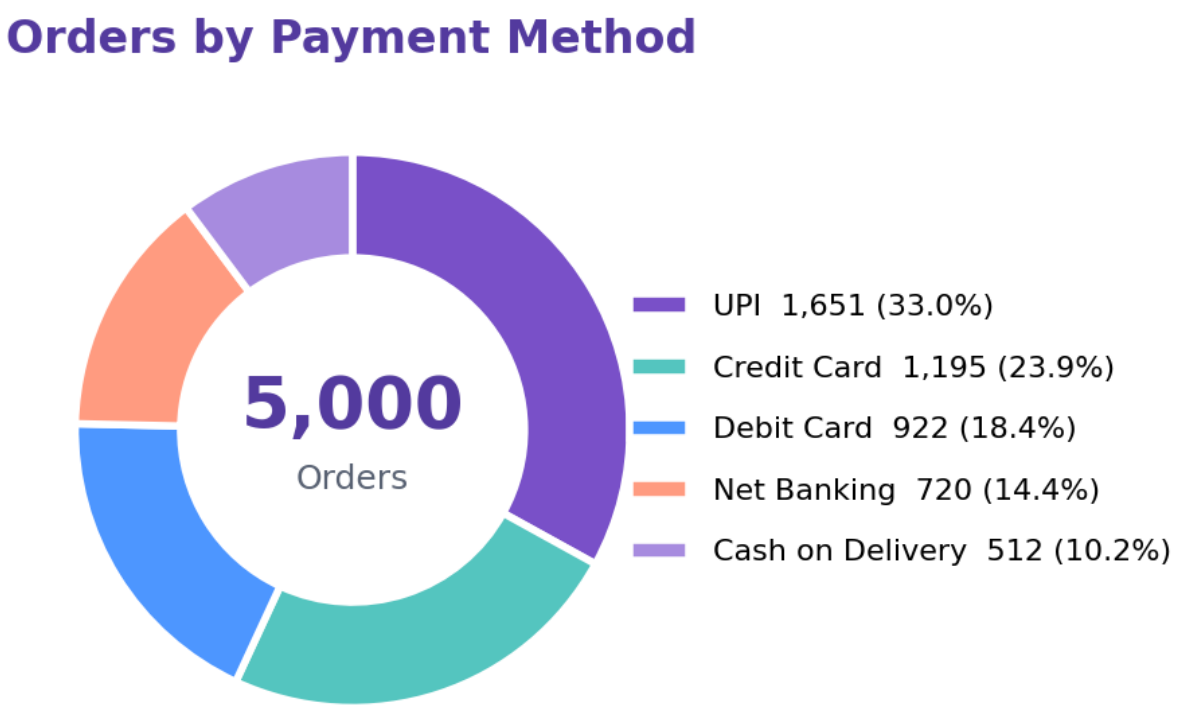


Figure 4. Orders by payment method.

Payment method	Orders	Share of orders
UPI	1,651	33.02%
Credit Card	1,195	23.90%
Debit Card	922	18.44%
Net Banking	720	14.40%
Cash on Delivery	512	10.24%

Observation: UPI was the most frequently used payment method with 1,651 orders (33.02%), followed by Credit Card with 1,195 orders (23.90%). Cash on Delivery was the least used with 512 orders (10.24%).

6.6 Top Customers by Purchases

The bar chart ranks customer names by the number of order records associated with each name. If multiple CustomerIDs share a name, a visual grouped by CustomerName combines those records; CustomerID is the safer identifier for unique-customer analysis.

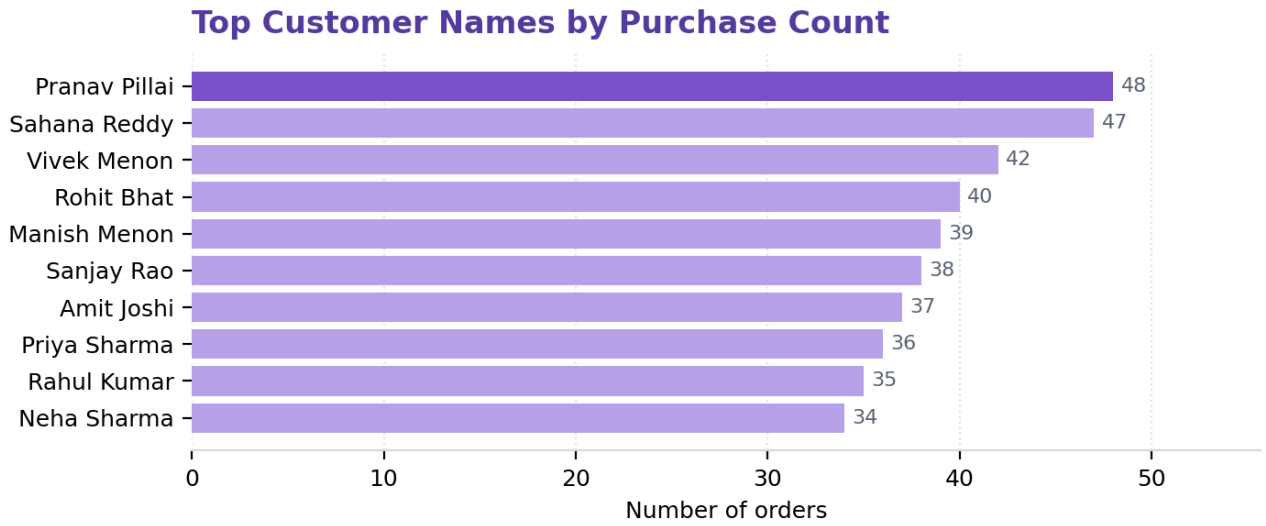


Figure 5. Top customer names by purchase count.

Customer name	Purchases	Revenue
Pranav Pillai	48	459,417.00
Sahana Reddy	47	466,641.50
Vivek Menon	42	304,742.00
Rohit Bhat	40	522,152.00
Manish Menon	39	641,738.00
Sanjay Rao	38	675,635.50
Amit Joshi	37	458,019.50
Priya Sharma	36	133,631.00
Rahul Kumar	35	244,568.00
Neha Sharma	34	390,231.00

6.7 Customer Revenue and Purchase Analysis

The customer table in Power BI displays customer name, total revenue and total orders. It supports comparison of purchasing frequency with sales contribution.

6.8 Revenue by Customer Segment

Customer segment	Customer profiles	Revenue	Revenue share
High Value	184	29.45M	62.60%
Medium Value	366	15.40M	32.73%
Low Value	450	2.19M	4.67%

The High Value segment contributes the largest share of revenue. Maintaining satisfaction and retention among high-value customers should therefore be a business priority.

7. Answers to the Assignment Questions

1. Which product category is most popular?

Electronics is the most popular category by revenue, generating 37.48M from 1,095 orders and contributing 79.67% of total revenue.

2. Which payment method is most frequently used?

UPI is the most frequently used method, with 1,651 orders, equal to 33.02% of all orders.

3. How does revenue change month by month?

Revenue fluctuates through the year. August is the peak month at 4.66M, while June is the lowest at 3.37M; average monthly revenue is approximately 3.92M.

4. Which customers have made the most purchases?

When grouped by CustomerName, the leading names are Pranav Pillai (48), Sahana Reddy (47), Vivek Menon (42), Rohit Bhat (40), Manish Menon (39). The chart can combine customers that share a name, so CustomerID is recommended for precise unique-customer counts.

5. Are a small number of customers responsible for a large proportion of revenue?

Not the top 10 alone: the top 10 CustomerIDs generate 4.15M, approximately 8.81% of total revenue. Still, the top 20% of active customers generate about 62.43%, so high-value customers remain important.

8. Key Business Insights

- Total recorded sales were 47.04M across 5,000 orders; 916 customer profiles placed at least one order and average order value was 9,407.92.
- Electronics contributes nearly 80% of revenue and has the most orders (1,095), while UPI is the leading payment method at 33.02% of orders.
- Monthly revenue ranged from 3.37M in June to 4.66M in August.
- The top 10 CustomerIDs contribute 8.81% of revenue, while the top 20% of active customers contribute 62.43%.

9. Limitations

- The dataset has no product-cost or expense fields, so actual profit and profit margin cannot be calculated.
- CustomerName is not unique. Name-based visuals may combine multiple CustomerIDs; use CustomerID for unique-customer analysis.
- The dataset covers only 2025, and the Top 10 Customer Revenue measure changes with slicer selections.

10. Conclusion

The Customer Analysis Dashboard was developed in Power BI using order and customer datasets, KPI cards, slicers, bar charts, a line chart, a donut chart, and a customer analysis table. The analysis shows that Electronics is the leading product category, UPI is the most frequently used payment method, and monthly revenue is variable but broadly stable over the year. Revenue is not dominated by the top ten customers alone, although the highest-value customer group contributes a substantial portion of sales. Cost information was unavailable, so the report focuses on revenue rather than profit.

11. Learning Outcomes

- Imported and modelled customer and order datasets and created a relationship using CustomerID.
- Created DAX measures for revenue, orders, active customers, average order value and top-N revenue.
- Designed interactive cards, charts, tables and slicers to explore customer and sales behaviour.
- Interpreted business patterns and identified limitations such as missing cost data and duplicate customer names.